

# STAT 5200, Notes 7<sup>1</sup>

Minchul Shin<sup>2</sup>

Fall 2025

**Readings.** Read Section 4.4 in W. It deals with the bias in OLS estimation when there is measurement error. Measurement error can be in the response variable, and it can be in the explanatory variables. We'll look at both of these situations. In the next set of notes we'll start to study instrumental variables estimation. Start reading Chapter 5 in W.

**Measurement error in the response variable.** Suppose  $y^*$  is the desired response variable. The regression model we want to consider is

$$y^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + v, \quad (1)$$

and we assume  $v$  has mean 0 and is uncorrelated with the  $x_i$ 's. We are really interested in  $E[y^*|x_1, \dots, x_K]$ . Let's assume that  $E[y^*|x_1, \dots, x_K] = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K$ . Instead of observing  $y^*$ , however, we observe  $y$ , and  $e_0 = y - y^*$  is the measurement error. Thus, the structure we actually estimate is

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + v + e_0. \quad (2)$$

We assume the measurement error has mean zero; if not, we simply adjust the intercept in (2). It is common to assume that  $e_0$  and  $v$  are uncorrelated.

Suppose the dependent variable is specified on the log scale. Then it is natural to assume the measurement error arises multiplicatively, that is,  $y = y^* a_0$ , so that  $\log y = \log y^* + e_0$ , where  $e_0 = \log a_0$ .

**Case 1:**  $e_0$  is uncorrelated with the explanatory variables. If the measurement error has zero correlation with all of the explanatory variables, and we employ (2) for estimation, OLS estimation will produce consistent estimation of the partial slopes and the intercept. The usual OLS tests and confidence regions will be valid. In this case, the only impact of the measurement error in the dependent variable is to inflate the error variance, which is

$$\text{Var}(v + e_0) = \text{Var}(v) + \text{Var}(e_0) > \text{Var}(v).$$

That is, standard errors will be larger, deflating  $t$  and  $F$  ratios (or,  $z$  and Wald ratios), and confidence intervals will be wider, than in the setting with no measurement error.

---

<sup>1</sup>This is based on Paul Shaman's lecture notes (Notes 7 from STAT 5200) at the University of Pennsylvania. Last revised: Fall, 2025. Last compiled: November 10, 2025. Lecture notes and associated materials are licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license.

<sup>2</sup>Email: [mincshin@wharton.upenn.edu](mailto:mincshin@wharton.upenn.edu)

**Case 2:**  $e_0$  is correlated with the explanatory variables. In some situations one has to assume that the measurement error in the dependent variable is correlated with the independent variables. This will lead to bias in OLS estimation. To understand the bias, we form the linear projection  $L(e_0|1, x_1, \dots, x_K)$ ; that is, we write

$$e_0 = \rho_0 + \rho_1 x_1 + \dots + \rho_K x_K + r, \quad (3)$$

where  $r$  has mean zero and is uncorrelated with the  $x_i$ 's. Now substitute (3) into (2), yielding

$$y = (\beta_0 + \rho_0) + (\beta_1 + \rho_1)x_1 + \dots + (\beta_K + \rho_K)x_K + v + r. \quad (4)$$

Thus, in this case, the OLS estimates from (2) of the partial slopes and the intercept will be biased. In some applications, it may be reasonable to assume some of the  $\rho_i$ 's in (3) are zero, and then consistent estimation of the corresponding  $\beta_i$ 's will occur with OLS.

**Measurement error in the explanatory variables.** Let's consider a single explanatory variable measured with error. Let  $x_K^*$  be the explanatory variable which is contaminated by additive error. It is unobserved and is often referred to as the latent variable. The model we want to address is

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_K x_K^* + v, \quad (5)$$

where  $v$  has mean 0 and is uncorrelated with the  $x_i$ 's,  $i = 1, \dots, K-1$ , and uncorrelated with  $x_K^*$ . We are really interested in  $E[y|x_1, \dots, x_K^*]$ . Let's assume that  $E[y|x_1, \dots, x_K^*] = \beta_0 + \beta_1 x_1 + \dots + \beta_K x_K^*$ . Instead of observing  $x_K^*$ , we observe  $x_K$ , and

$$e_K = x_K - x_K^* \quad (6)$$

is the measurement error. Thus, the structure we actually estimate is

$$\begin{aligned} y &= \beta_0 + \beta_1 x_1 + \dots + \beta_K x_K + (v - \beta_K e_K) \\ &= \beta_0 + \beta_1 x_1 + \dots + \beta_K x_K + u. \end{aligned} \quad (7)$$

The error term in (7), the structure we estimate, is  $u = v - \beta_K e_K$ .

We can assume that  $e_K$  has mean zero, because we include an intercept in (7). Thus,  $v - \beta_K e_K$  has mean 0. Furthermore, it is customary and not controversial to assume

$$Cov(x_k, e_K) = 0, \quad k = 1, \dots, K-1. \quad (8)$$

We also need to understand how the error term  $v - \beta_K e_K$  is correlated with the  $x_i$ 's. First, let's note that we make a redundancy assumption,

$$E[y|x_1, \dots, x_{K-1}, x_K^*, x_K] = E[y|x_1, \dots, x_{K-1}, x_K^*]. \quad (9)$$

This assumption states that  $x_K$  has no effect on  $y$  after removal of the collective effect of  $x_1, \dots, x_{K-1}, x_K^*$ . It's certainly reasonable. The error  $v$  is uncorrelated with  $x_1, \dots, x_{K-1}, x_K^*$ , and by the redundancy assumption (9),

$$\text{Cov}(x_K, v) = 0 \quad (10)$$

(recall Property CE.6 in W). The *classical errors-in-variables* (CEV) assumption in the literature is that the measurement error is uncorrelated with the unobserved explanatory variable,

$$\text{Cov}(x_K^*, e_K) = 0. \quad (11)$$

Let's make this assumption (is it reasonable? maybe not? certainly not for some choices of  $x_K^*$ ). From (6) and (11) it follows that

$$\text{Cov}(x_K, e_K) = \text{Cov}(x_K^* + e_K, e_K) = \text{Var}(e_K). \quad (12)$$

Then, by (10) and (12), the covariance between  $x_K$  and the error term in the regression we are fitting,  $v - \beta_K e_K$ , is

$$\text{Cov}(x_K, v - \beta_K e_K) = -\beta_K \text{Var}(e_K).$$

Therefore, under the CEV assumption, OLS estimation of model (7) will yield biased (inconsistent) estimates of all the  $\beta_i$ 's. We can characterize the bias of the estimator of  $\beta_K$  in (7) under the given assumptions.

As Wooldridge notes, it is difficult to give a clear characterization of the biases of the estimators of the other  $\beta_i$ 's in (7). This is because it is difficult to analytically characterize  $L(e_K|1, x_1, \dots, x_K)$ .

To characterize the bias of the estimator of  $\beta_K$  when we use (7) with OLS, we apply the steps in Problem 2.6 of W, followed by (2.56) in Property LP.7. See problem 4.10 in W. First we show that two linear projections are identical (but, careful – the residuals are different):

$$\begin{aligned} L(x_K|1, x_1, \dots, x_{K-1}) &= L(x_K^* + e_K|1, x_1, \dots, x_{K-1}) \\ &= L(x_K^*|1, x_1, \dots, x_{K-1}) + L(e_K|1, x_1, \dots, x_{K-1}) \\ &= L(x_K^*|1, x_1, \dots, x_{K-1}), \end{aligned} \quad (13)$$

where the last step follows from (8) and  $E[e_K] = 0$ . Next, we write the two projections in (13) with their residuals:

$$x_K = L(x_K|1, x_1, \dots, x_{K-1}) + r_K, \quad (14)$$

$$x_K^* = L(x_K^*|1, x_1, \dots, x_{K-1}) + r_K^*. \quad (15)$$

By (6) and (13)–(15) we can write

$$\begin{aligned} x_K &= x_K^* + e_K \\ &= L(x_K^*|1, x_1, \dots, x_{K-1}) + r_K^* + e_K \\ &= L(x_K|1, x_1, \dots, x_{K-1}) + r_K^* + e_K \\ &= L(x_K|1, x_1, \dots, x_{K-1}) + r_K. \end{aligned} \quad (16)$$

That is, the linear projection of  $x_K$  onto  $1, x_1, \dots, x_{K-1}$  has residual  $r_K = r_K^* + e_K$ . This is just the result of part b of Problem 2.6 in W. Now apply (2.56) of Property LP.7 and (2.39) in W to express the probability limit of the estimator of  $\beta_K$  in the regression (7) (see line 12 on page 13 of Notes 1):

$$\text{plim}(\hat{\beta}_K) = \frac{\text{Cov}(y, r_K)}{\text{Var}(r_K)} = \frac{\text{Cov}(y, r_K^* + e_K)}{\text{Var}(r_K^* + e_K)} = \frac{\text{Cov}(y, r_K^*) + \text{Cov}(y, e_K)}{\text{Var}(r_K^* + e_K)}. \quad (17)$$

Consider  $\text{Cov}(y, r_K^*)$  in the numerator of (17) and note  $y$  is given by (5). From (15)  $r_K^*$  has mean 0 and is uncorrelated with  $x_1, \dots, x_{K-1}$ . Since  $v$  is uncorrelated with  $x_K^*$ , (15) implies  $v$  is also uncorrelated with  $r_K^*$ . Thus, we can write, using (5) and (15),

$$\text{Cov}(y, r_K^*) = \beta_K \text{Cov}(x_K^*, r_K^*) = \beta_K \text{Var}(r_K^*). \quad (18)$$

Next consider  $\text{Cov}(y, e_K)$  in the numerator of (17). It is

$$\text{Cov}(\beta_0 + \beta_1 x_1 + \dots + \beta_K x_K^* + v, e_K),$$

and this is 0 by (5), (6), (8), (10) and (11). Thus, the numerator of (17) is given by (18).

Next, the denominator of (17) is

$$\text{Var}(r_K^*) + \text{Var}(e_K) + 2\text{Cov}(r_K^*, e_K).$$

The covariance is 0 by (8), (11) and (15). Therefore, the probability limit of  $\hat{\beta}_K$  is

$$\beta_K \frac{\text{Var}(r_K^*)}{\text{Var}(r_K^*) + \text{Var}(e_K)}. \quad (19)$$

This is a relatively simple formula. The ratio in (19) gives the attenuation bias arising from OLS estimation of (7) and attributable to the error  $e_K$  in observation of the explanatory variable  $x_K^*$ .<sup>3</sup> The attenuation bias becomes more severe as the variance of  $r_K^*$  decreases, that is, as the multiple correlation between the unobserved variable  $x_K^*$  and the other covariates in (5),  $x_1, \dots, x_{K-1}$ , increases (as this collinearity increases).

The above discussion assumes only one of the explanatory variables has been measured with error. The situation is very messy if more than one is measured with error and is hard to characterize.

In some situations we cannot make the CEV assumption that the measurement error is uncorrelated with the unobserved explanatory variable. For example, if  $x_K^*$  is a binary variable, the measurement error obviously depends on the true value.

---

<sup>3</sup>In the special case  $K = 1$ , (19) becomes

$$\beta_1 \frac{\text{Var}(x_1^*)}{\text{Var}(x_1^*) + \text{Var}(e_1)}. \quad (20)$$

The assumptions and the result described above do not constitute the only approach to study measurement error in an explanatory variable which has been considered in the literature. It has been shown that random misclassification of a binary explanatory variable biases a regression coefficient toward zero. There is a substantial errors-in-variables literature.